



SECURITY CONCEPT

Dr. Sanjay Adiwaj

Scientist E

C-DAC Bangalore



सी डैक
CDAC

Agenda

- Facts
- Internet
- Internet Threats – Social Engineering
 - Demos
 - Real World Examples - Phishing
 - Incidents @C-DAC
- Privacy Concerns using AI
- Reverse Image Search
- Privacy Concerns on LinkedIn
- Digital Arrest



7.98 billion people in the world, according to the latest UN figures, the study counted 5 billion users connected to the internet.



of which 92.4% use their phone to get online.



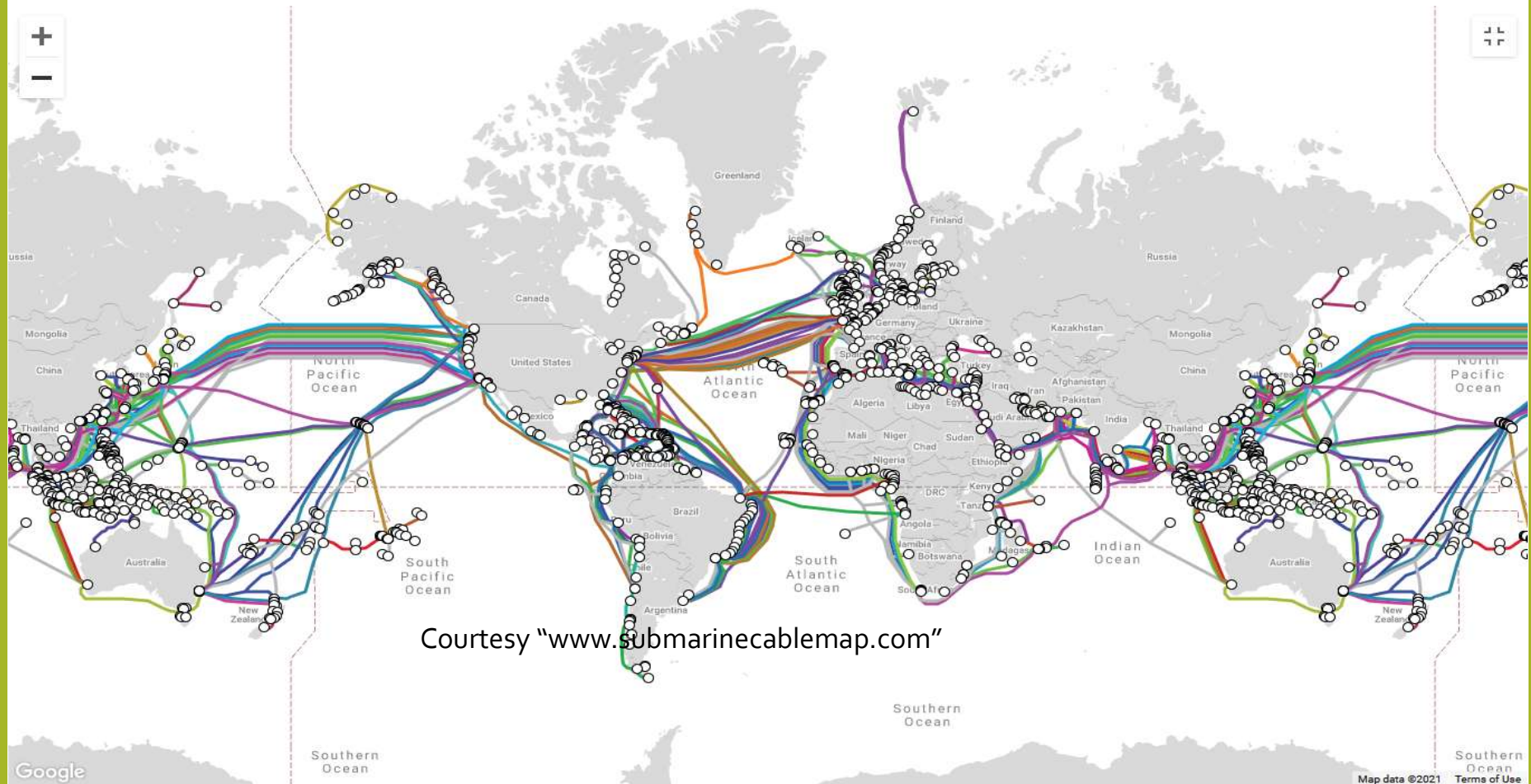
On average, an internet user spends
6 hours, 53 minutes online per day.



- **Internet** (short for interconnected network)
- Global system of interconnected computer networks
- Internet protocol suite (TCP/IP)



The Internet on MAP



Courtesy "www.submarinecablemap.com"

Facts

- Every 39 seconds there is a cyber attack
- 4 million files are stolen every day - that's 44 every single second
- Facebook knows when you are falling in love
- More people have access to the internet than clean drinking water.

Courtesy - <https://www.dbxuk.com/statistics/cyber-security>



How and for what purpose do you use the Internet ?

Education



Communication

E-mail



Online Shopping



Online Banking



Social Networking



Fun/Entertainment

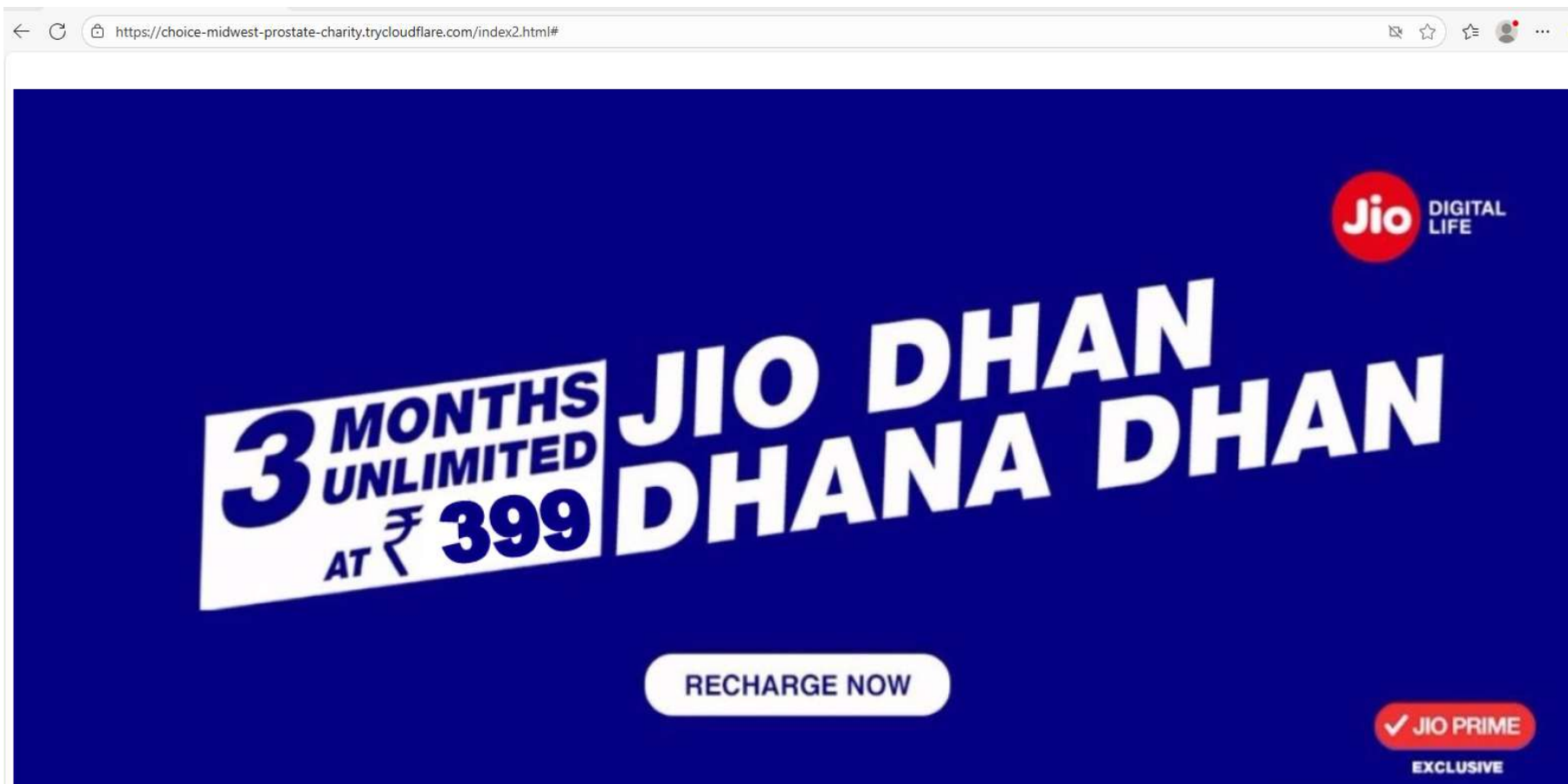


Ask your self whether Internet usage is safe or not ??



Demo

- Never Trust any Website/Link
- Consequences of Clicking on Links



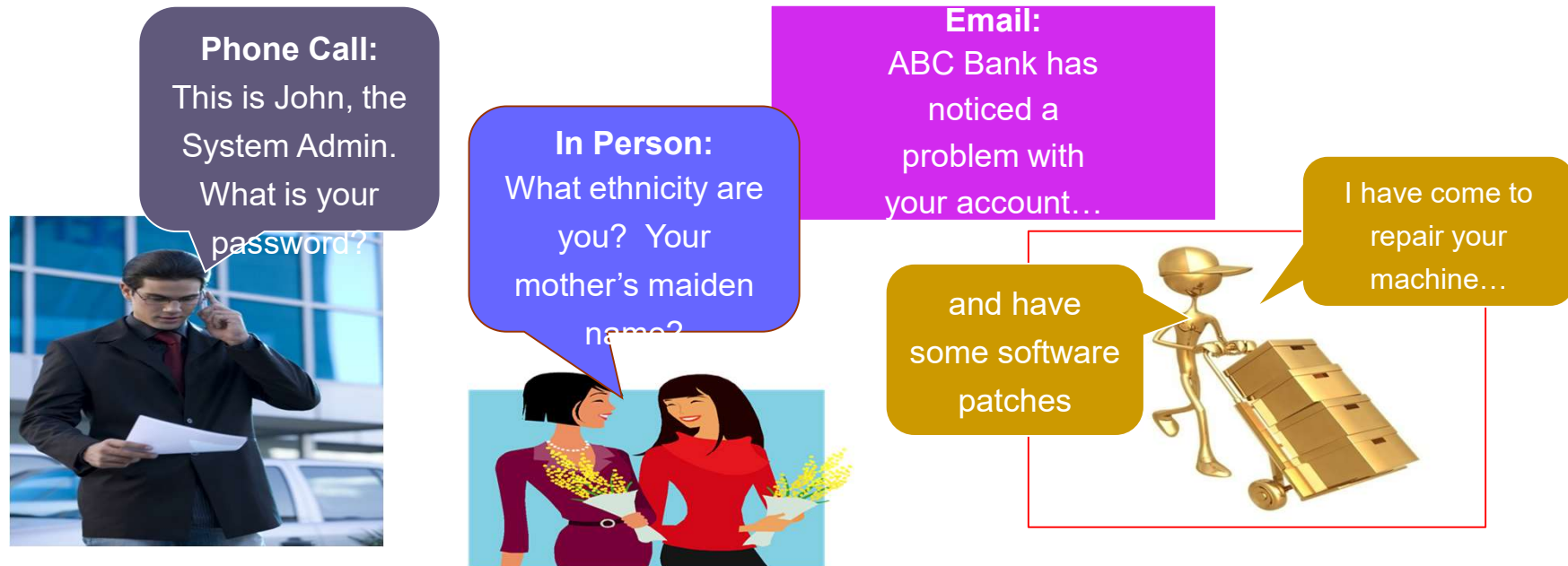
Why is Cyber Security crucial today

- Which is the Third Largest economy?
 - US
 - China
 - ?
- Global Cybercrime Damage Costs (Cyber Security Ventures)
 - \$6 Trillion USD a Year
 - \$500 Billion a Month
 - \$115 Billion a Week
 - \$16 Billion a Day
 - \$11.4 Million a Minute

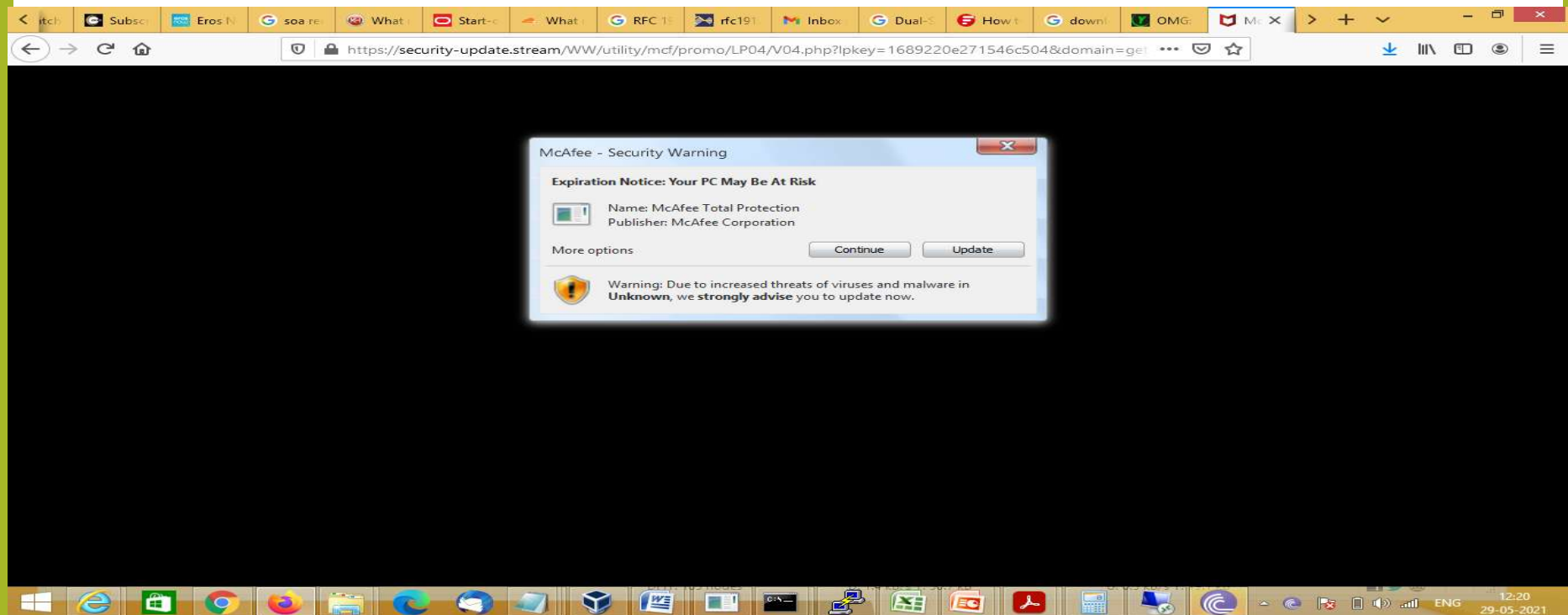


LEADING THREATS - SOCIAL ENGINEERING

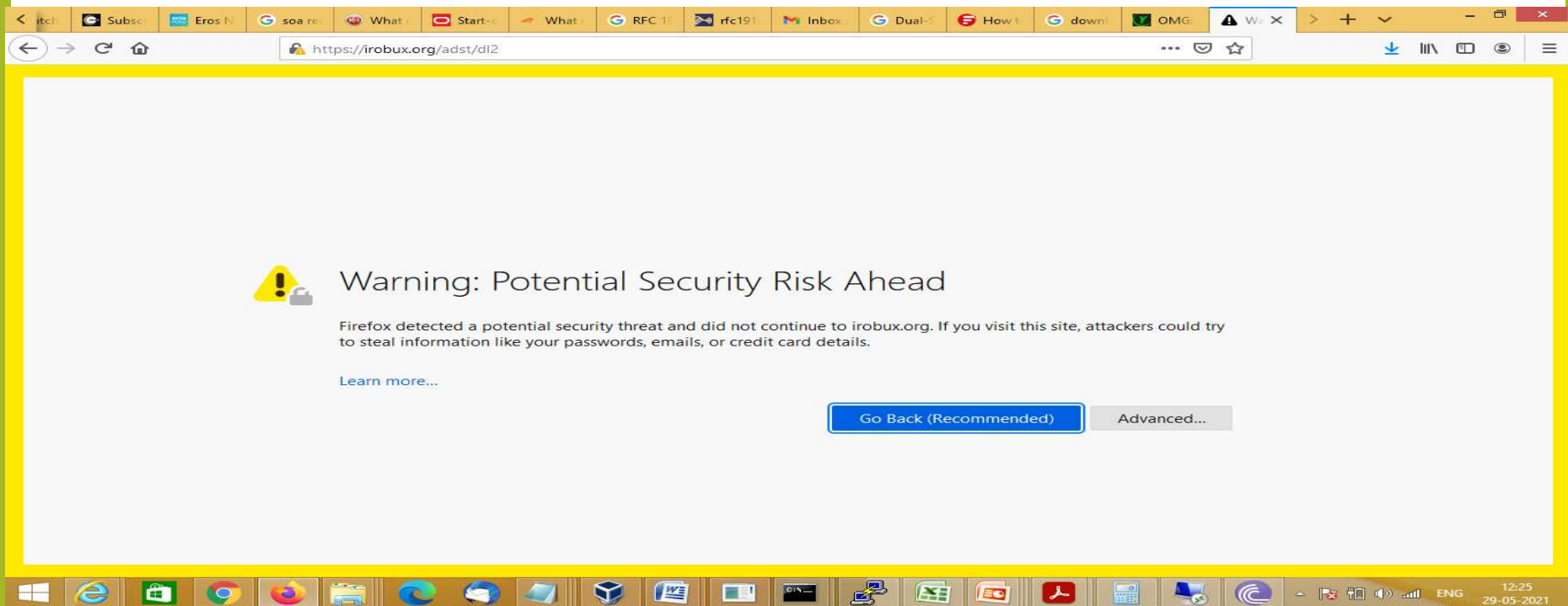
- ▶ Social engineering manipulates people into performing actions or divulging confidential information. Similar to a confidence trick or simple fraud, the term applies to the use of deception to gain information, commit fraud, or access computer systems.



SOCIAL ENGINEERING – Through Web



SOCIAL ENGINEERING – Through Web



Email-Based Social Engineering

- Most common form: **Phishing**
- Emails designed to trick users into:
 - Clicking malicious links
 - Downloading infected attachments
 - Revealing credentials or sensitive info
- In 2025, C-DAC was targeted by multiple phishing email attacks.



Phishing Email

From: "cdac.in" <admin@cdac.in>
To: g...@cdac.in
Date: 01/15/2025 7:25 PM
Subject: @cdac.in Password Renewal

Password Credentials

Your password expires now, and [g...@cdac.in](#) will be closed automatically. Click below to continue with the same password.

Keep the same password

Webmaster for cdac.in Account and Services.
© 2025

Webmail

Email Address

Password

Log in

English العربية čeština dansk Deutsch Ελληνικά español español latinoamericano ...



Copyright © 2025 cPanel, L.L.C.
Privacy Policy

Phishing Email

From: Mail administrator <56983145746558@cdac.in>

To: [Redacted]

Subject: Action Required

Dear Sir/Mam,

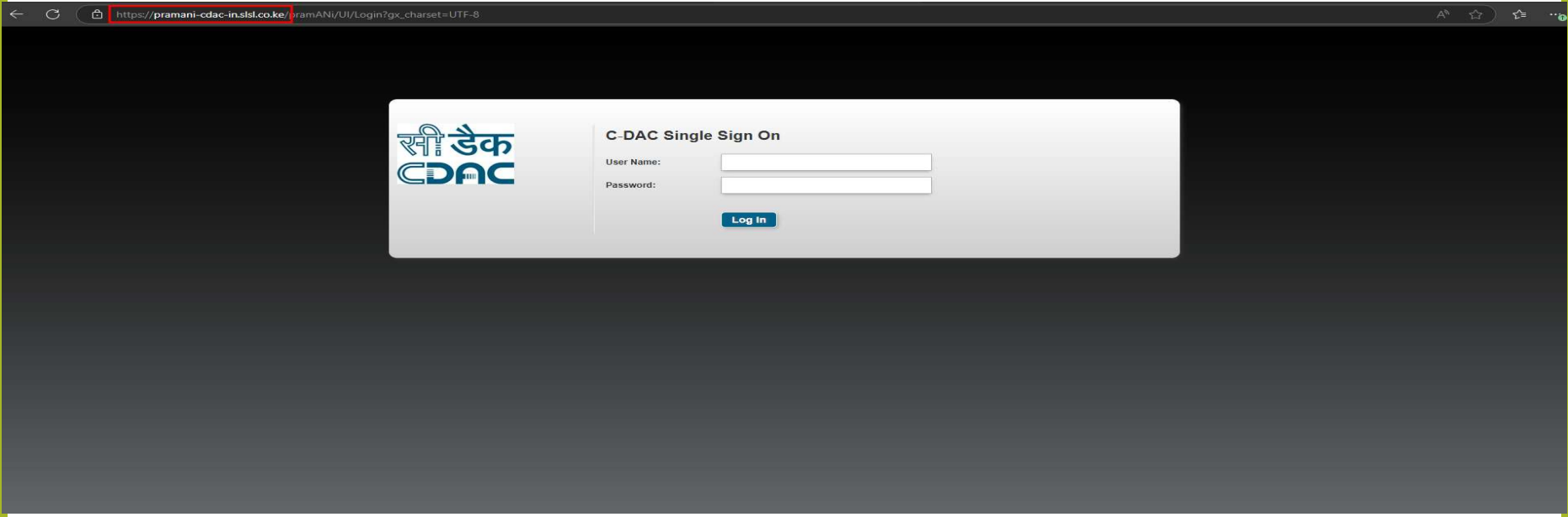
We are contacting you to request that you verify your email address and password associated with your account.

To complete this verification, please follow the steps below:

[Click here](#)

With Best Regards,

Mail Administrator



The screenshot shows a web browser window with a dark background. In the center, there is a white rectangular box containing the CDAC logo on the left and a login form on the right. The login form is titled "C-DAC Single Sign On" and includes fields for "User Name:" and "Password:". Below these fields is a blue "Log In" button. The browser's address bar shows a URL that has been modified to "https://pramani-cdac-in.slsf.co.ke/ramANI/UI/Login?gx_charset=UTF-8", with the domain part highlighted in red.

https://pramani-cdac-in.slsf.co.ke/ramANI/UI/Login?gx_charset=UTF-8

 C-DAC Single Sign On

User Name:

Password:

Phishing Email

----- Original Message -----

From: "cdac.in IT Support" <wren@traveleza.com>

To: [redacted]@cdac.in

Date: 04/04/2025 7:00 AM

Subject: Administrator on [redacted]@cdac.in

WEBMAIL PASSWORD VALIDATION

Your password is set to expire. To ensure uninterrupted access to your account, please update your password before the expiration date.

Account Information:

Email Address: [redacted]@cdac.in

Password Status: **About to Expire**

Keep My Password

Modify Password

Phishing Email

CDAC ACCOUNT

we may have prevented the delivery of 3 new messages
to your inbox as of **Today, 11:26am** due to your email storage capacity.
You can use the instruction below to free up some space.

Clear Cache

[Was this helpful?](#)

Thanks

Phishing Email

- Seems to be email from Ministry of Corporate Affairs

बहुत ज़रूरी।

○ **mca.gov.in** <denis.zelj@amis.net>
To undisclosed-recipients;

6/22/2025 7:28 AM

Reply Reply all Forward Delete

नमस्ते mca.gov.in उपयोगकर्ता,

हम सभी अप्रयुक्त mca.gov.in खाते बंद कर रहे हैं। आपकी ईमेल सेवा में व्यवधान से बचने के लिए, कृपया नीचे अपने mca.gov.in ईमेल खाते की पुष्टि करें।

पुष्टि करने के लिए यहाँ क्लिक करें

<https://santaluzia.ibsystem.com.br/premios/mail.mca.gov.in/mca.gov.in/>

नोट: यदि आप अनुपालन नहीं करते हैं, तो हम आपके खाते को लॉक कर देंगे और सभी ईमेल डेटा स्थायी रूप से खो देंगे।

धन्यवाद
mca.gov.in

Phishing Email

----- Original Message -----

From: linn chi-shun <lbangmesser01@sprintpcs.com>

To: @cdac.in

Date: 06/11/2025 11:02 AM

Subject: Re:

Good day.

Your system has been hacked with a trojan virus.

It has penetrated your device through adult portals which you sometimes visit.

Some spicy videos contain malicious code that activates after being turned on. Your entire information has already been copied to my servers.

I possess **complete control** over your device which you use to access the Internet.

I can see your screen, I can use a microphone and a camera in a way that you never notice anything.

I've already made a screen recording.

A video was edited with a **po**phic movie that you were watching at that time and r

Your face is perfectly visible and I don't think that this kind of content will have a positive impact on your reputation.

I have an overall access to your list of contacts and the social media profiles. I can send this video from your E-mail or the messengers.

If you don't want to let this happen, then you only need to take one simple step.

Just transfer **1300 USD** (US dollars) to Bitcoin wallet: 1C2HFpBi6b3FmwyXFpjc2G1rruXZKCFyZb

(In a Bitcoin equivalent at the exchange rate for the time of transfer)

You can find the detailed instructions in Google.

After the payment I will remove the video and the virus from your device and no one will bother you anymore.

If I won't receive the payment in due time, all of your data and the videos will become publicly available.

I give you 2 days.

I shall receive a notification that you have read the letter.

The timer starts immediately.

Any complain somewhere, including the police, is useless. My wallet and an E-mail cannot be tracked.

If I find out that you have shared this message with someone else, the video will become publicly available at once.

I will destroy your reputation forever and all your data will go public.

Everyone will learn about your passion for the **s**ites and more. Changing the passwords will be useless either as all the data is already on my servers.

Don't forget that reputation is very important and be prudent!

Phishing Email

From: "cdac.in" <esqnh@cdac.in>
To: @cdac.in
Date: 05/30/2025 9:42 AM
Subject: FW: WEB-PROJECT 010-T-0028,.

Password Credentials

Your password expires now, and @cdac.in will be closed automatically. Click below to continue with the same @cdac.in password.

[Keep the same password](#)

Webmaster for cdac.in
© 2025

Phishing Email

Re: C-DAC Bangalore  Inbox x



Dr S. D. Sudarsan <arkiwave@singnet.com.sg>
to me ▼

08:38 (1 hour ago)

Hi Balakrishna

Let me know if you are free presently? I need you to make some findings for me.

Thanks

Phishing Email

WEBMAIL PASSWORD VALIDATION

Your password is set to expire, To ensure uninterrupted access to your account, please update your password before the expiration date.

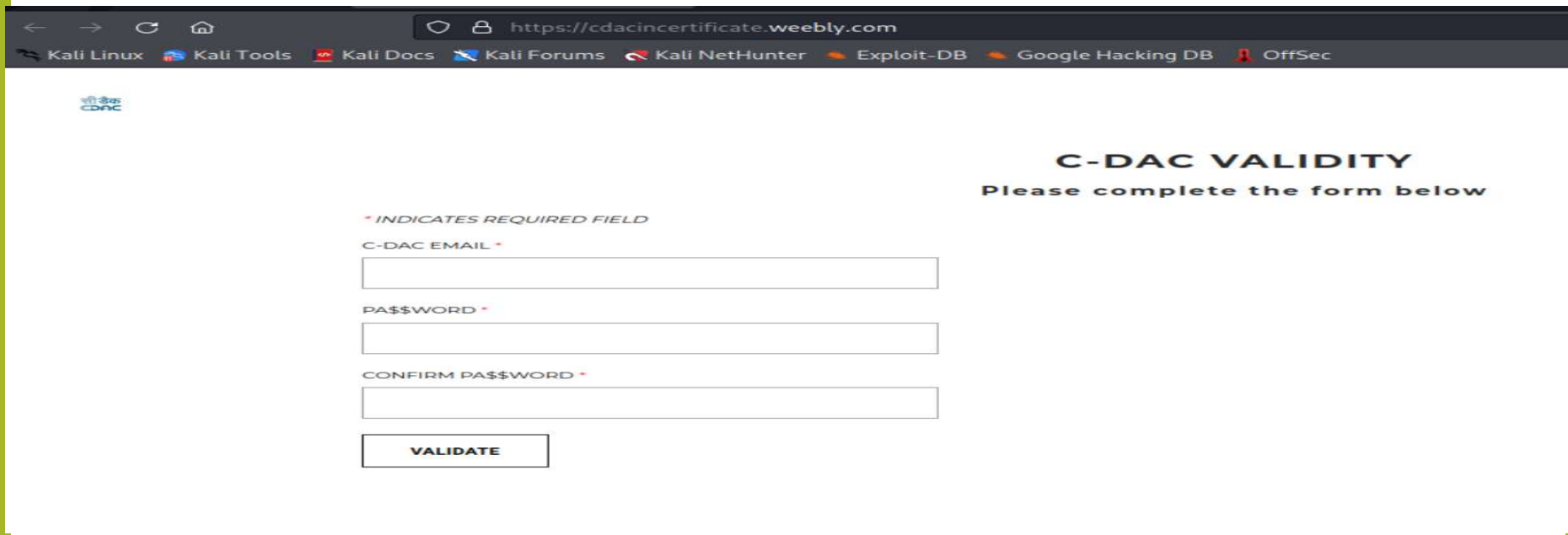
Account Information:

Email Address:

Password Status: **About to Expire**

Keep My Password

Modify Password



The screenshot shows a web browser window with the address bar displaying `https://cdacincertificate.weebly.com`. The browser's bookmark bar includes links to Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, Google Hacking DB, and OffSec. The website's header features the CDAC logo and the title "C-DAC VALIDITY" with the instruction "Please complete the form below". A note states "* INDICATES REQUIRED FIELD". The form contains three input fields: "C-DAC EMAIL *", "PA\$\$WORD *", and "CONFIRM PA\$\$WORD *". A "VALIDATE" button is positioned at the bottom of the form.



Demo - Phishing

← ↻ ⚠ Not secure 10.180.6.218



CDAC Single Sign On

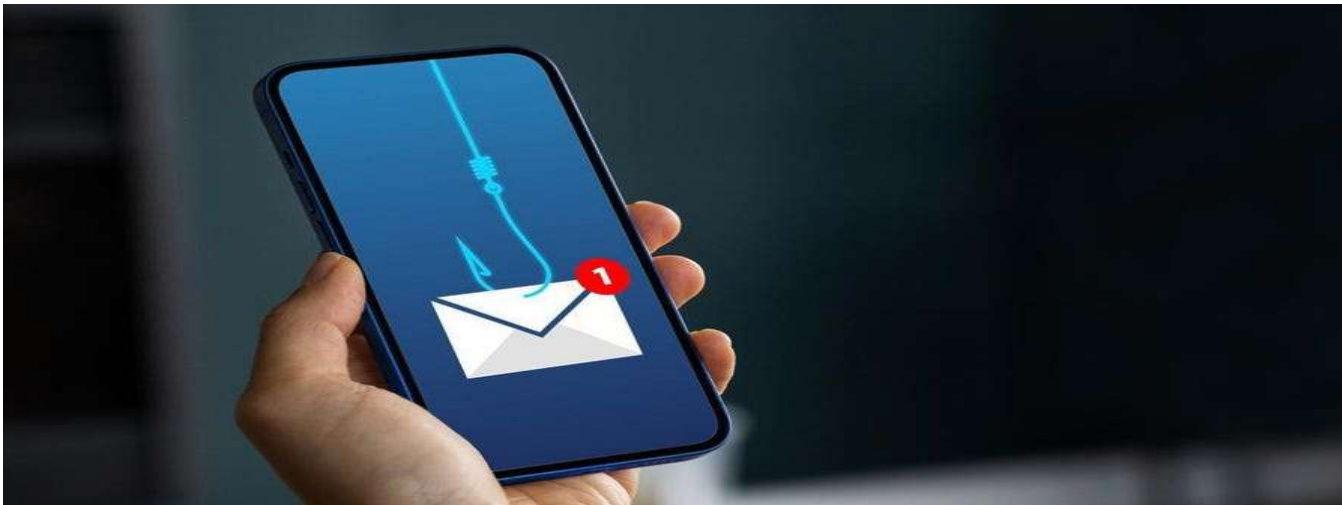
User Name / Emp ID:

Password:

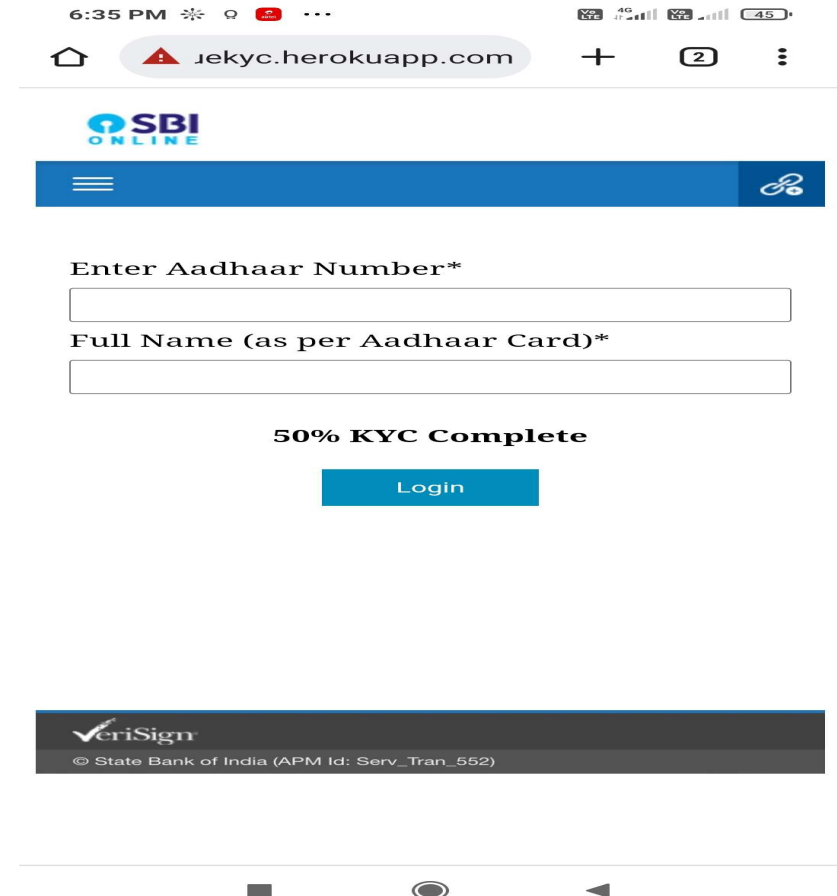
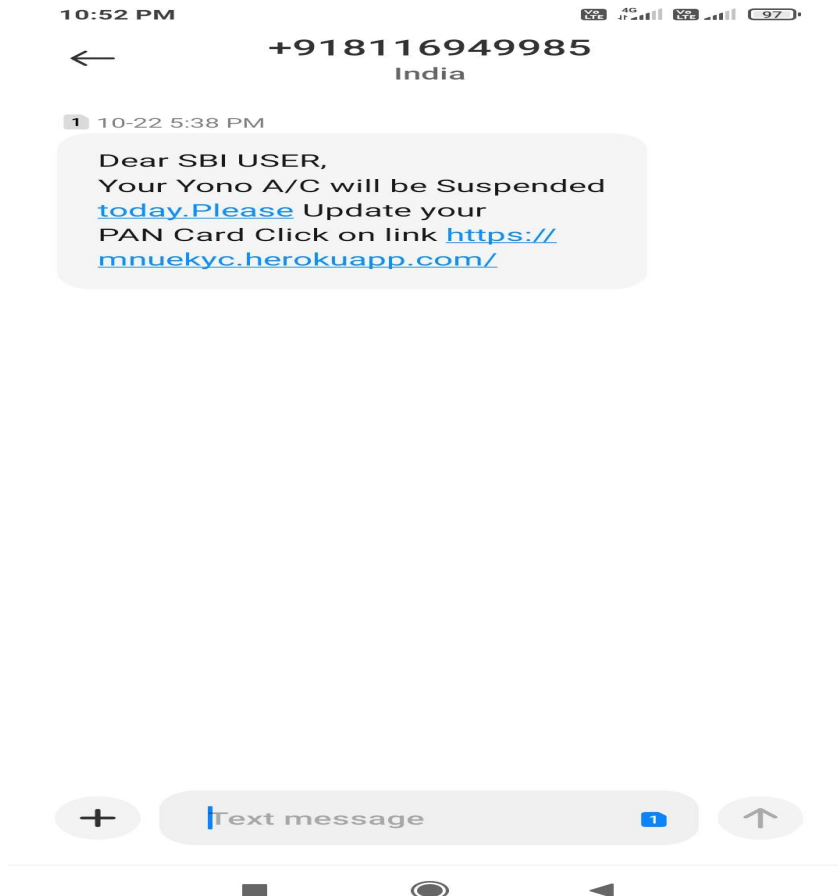
Log In

SOCIAL ENGINEERING – Through SMS

- **Smishing (SMS Phishing):** Attackers send fraudulent text messages containing malicious links or prompts to call a fake customer support number, aiming to trick recipients into revealing personal information or installing malware on their devices.



SMISHING EXAMPLE



SMISHING

6:35 PM 4G 45%

jekyc.herokuapp.com

SBI ONLINE **LOGOUT**

Welcome

Login Second Factor Authentication

Enter the One Time Password sent to your mobile phone

Mandatory fields are marked with an asterisk (*)

One Time Password(OTP)*

Wait for OTP : 115 secs

Login

If you did not receive the High Security Password on SMS, you can [Click here to resend the OTP](#)

- State Bank or any of its representative never sends you email/SMS or calls you over phone to get your personal information, password or one time SMS (high security) password. Any such e-mail/SMS or phone call is an attempt to fraudulently withdraw money from your account through Internet Banking. Never respond to such email/SMS or phone call. Please report immediately on report.phishing@sbi.co.in if you receive any such email/SMS or Phone call. Immediately change your passwords if you have

18/05/2026

6:34 PM 4G 45%

jekyc.herokuapp.com

SBI ONLINE

Enter PAN Card Number*

Full Name (as per PAN Card)*

Login

VeriSign
© State Bank of India (APM Id: Serv_Trans_552)

SMISHING

6:34 PM 4G LTE 45%

jekyc.herokuapp.com

SBI ONLINE **LOGOUT**

Welcome

Login Second Factor Authentication

Enter the One Time Password sent to your mobile phone

Mandatory fields are marked with an asterisk (*)

One Time Password(OTP)*

Wait for OTP : 108 secs

Login

If you did not receive the High Security Password on SMS, you can [Click here to resend the OTP](#)

- State Bank or any of its representative never sends you email/SMS or calls you over phone to get your personal information, password or one time SMS (high security) password. Any such e-mail/SMS or phone call is an attempt to fraudulently withdraw money from your account through Internet Banking. Never respond to such email/SMS or phone call. Please report immediately on report.phishing@sbi.co.in if you receive any such email/SMS or Phone call. Immediately change your passwords if you have

18/05/2026

6:33 PM 4G LTE 46%

jekyc.herokuapp.com

SBI ONLINE

Branch Name*

Account Holder Name*

Date Of Birth (DD/MM/YYYY)*

Login

VeriSign
© State Bank of India (APM Id: Serv_Tran_552)

SMISHING

6:32 PM 4G 46

jekyc.herokuapp.com

SBI **LOGOUT**

Welcome

Login Second Factor Authentication

Enter the One Time Password sent to your mobile phone

Mandatory fields are marked with an asterisk (*)

One Time Password(OTP)*

Wait for OTP : 50 secs

Login

If you did not receive the High Security Password on SMS, you can [Click here to resend the OTP](#)

- State Bank or any of its representative never sends you email/SMS or calls you over phone to get your personal information, password or one time SMS (high security) password. Any such e-mail/SMS or phone call is an attempt to fraudulently withdraw money from your account through Internet Banking. Never respond to such email/SMS or phone call. Please report immediately on report.phishing@sbi.co.in if you receive any such email/SMS or Phone call. Immediately change your passwords if you have

18/05/2026

Privacy concerns while using AI

- Ghibli-style filters
- The 130 million “users who have generated images”



- Sari/saree App
- because your photos contain biometric data and other info that can be reused, sold, or used to train models or create deepfakes.

How a Photo Contains Biometric Data

- Facial Features (Face Biometrics)
 - Details like the distance between your eyes, nose shape, jawline, mouth, facial symmetry, etc.
 - Measured and turned into a numerical template using facial recognition algorithms.
 - Basis of how systems like Face ID, airport security cameras, or surveillance systems identify people.



How a Photo Contains Biometric Data

- Iris and Eye Patterns
 - High-resolution images can capture detailed patterns in your irises, which are unique to each individual—even among identical twins.
 - Eye recognition systems use these patterns for biometric authentication.
- Skin Texture and Wrinkles:
 - Advanced systems may analyze skin texture, wrinkles, or moles to improve identification accuracy.
- Other Visible Traits:
 - Ears, scars, tattoos, or even the shape of your head and hairline can be used in some biometric systems.



Google Gemini's Nano Banana viral AI saree trend goes wrong

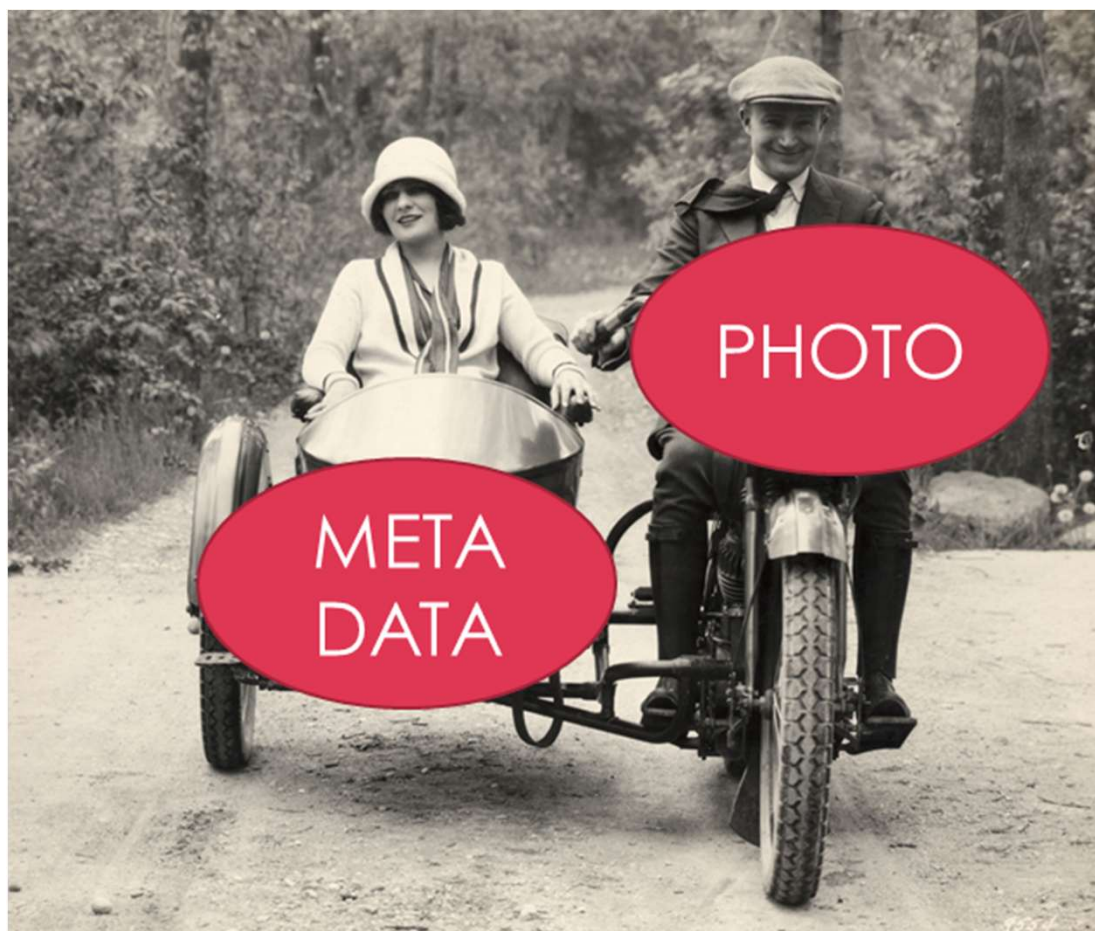


Reverse Image Search

- **Google Images / Google Lens** — upload an image or paste an image URL; Google shows matching images and pages. (How to use).
- **TinEye** — classic reverse image engine; you can upload an image, paste a URL, or drag & drop. It focuses on exact/modified copies.
- **Bing Visual Search** — upload/paste image or use camera icon in Bing Images; good alternative to Google.
- **Yandex Images** — often finds matches that Western engines miss; upload an image to search.

Metadata of Photos/Images

- Camera Make & Model
- Lens Info
- Shutter Speed
- Flash Used
- Date/Time Taken
- GPS Coordinates



Tool to analyze and display the metadata

→ exifinfo.org

Exif Info.org

An online tool to analyze and display the metadata in images and other media files.

Analyze a File:

Choose File No file chosen

Upload File

Or just drag and drop a file here.

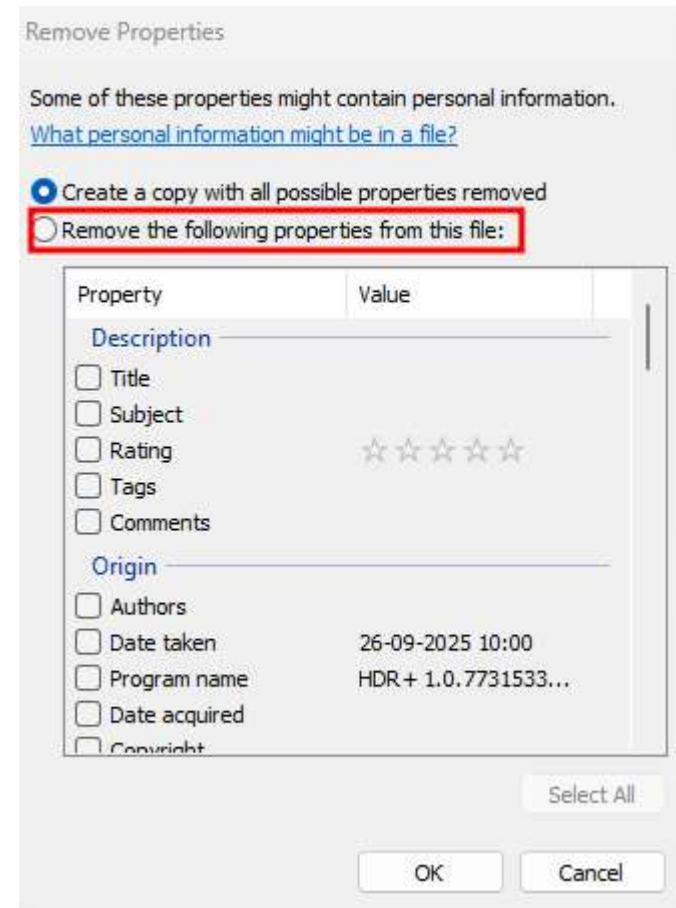
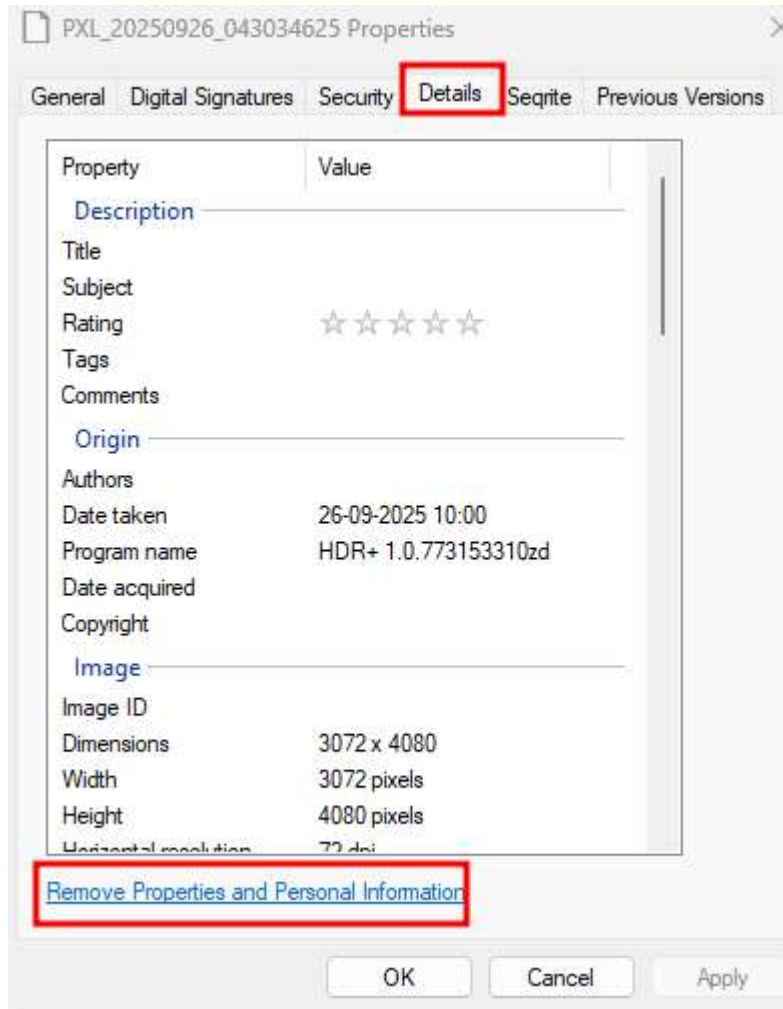
Analyze a URL:

URL:

Inspect URL

Enter the URL of a media asset.

How to Remove Metadata from an Image?



How to Reduce the Risk?

- **Don't upload photos you wouldn't want public** (especially ID photos, children, sensitive tattoos, or intimate images).
- **Prefer apps that process locally** (on-device) rather than uploading to cloud servers — check the app description/privacy policy for “on-device” or “local processing.”
- **Read the privacy policy & terms** (look for clauses about retention, training, resale, and deletion). If policy absent or vague, be cautious.
- **Strip metadata** before uploading (remove EXIF/GPS and use a copy): e.g., on phone or with an EXIF stripper.
- **Use throwaway images or avatars** for trends (or edit a low-resolution/cropped version).
- **Create an account using limited permissions** (don't connect social logins; don't give unnecessary app permissions).
- **Revoke/delete images** and accounts after use — check whether the app provides a deletion option and follow its deletion procedure. If you can't find deletion, contact their support and keep screenshots of requests.
- **Check app reputation**: app store reviews, number of installs, developer website, and news stories about misuse. Avoid unknown/clone apps.
- **Keep originals private** — don't reuse the same high-quality original across many services.

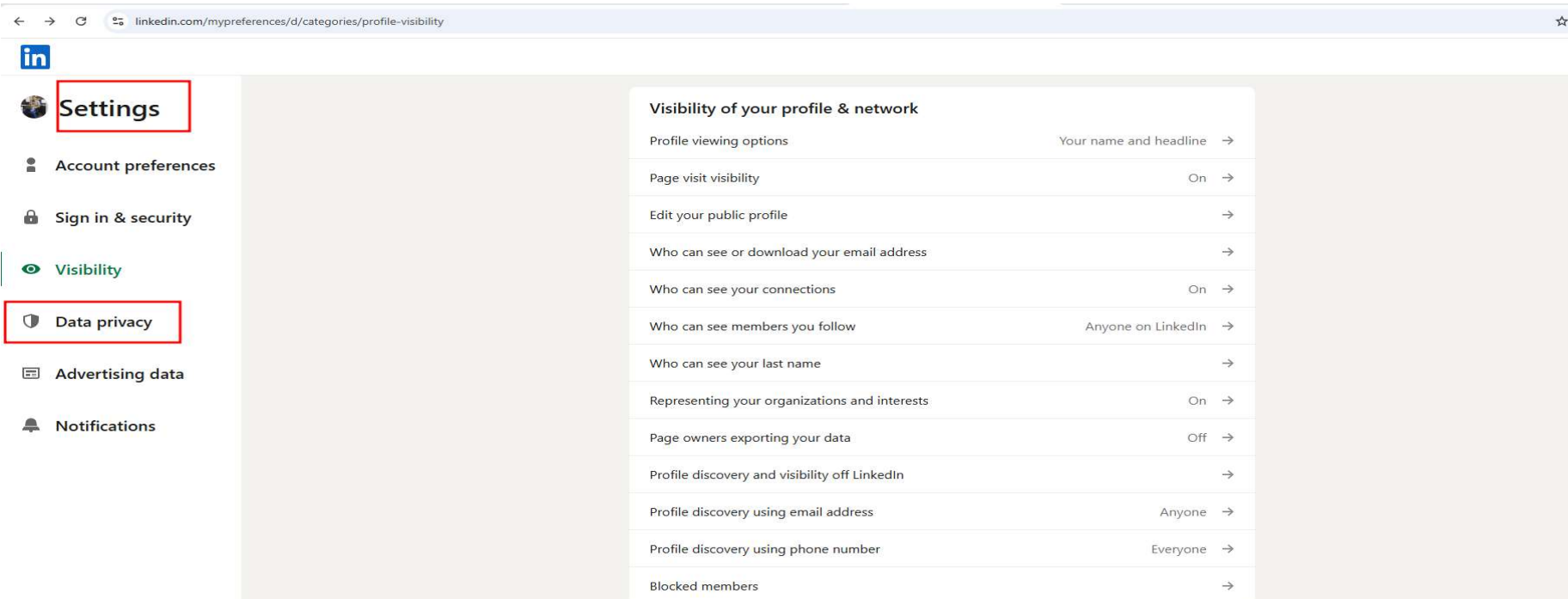
Privacy concerns on LinkedIn

- Exposure of Personal Information
 - Issue: Your full name, job history, education, location, and even contacts can be publicly visible.
 - Risk: Identity theft, social engineering attacks, and unsolicited contact.
- Third-Party App Access
 - Issue: Connecting LinkedIn to other platforms (job boards, resume builders, etc.) can leak data.
 - Risk: External platforms may misuse your profile information.
- LinkedIn profile could be soon used to train their AI Model?
 - This means that your public profile, posts, likes, comments and activity data will be used to train the AI models by default, UNLESS you opt out.



How to Protect Your Privacy on LinkedIn?

- Tighten Your Profile Visibility
- Go to **Settings > Visibility**
 - Set your profile to be visible only to your connections or “LinkedIn members.”
 - Control who can see your email address, connections, and last name.



The screenshot shows the LinkedIn 'Settings' page, specifically the 'Visibility' section. The left sidebar contains the following menu items: Settings (highlighted with a red box), Account preferences, Sign in & security, Visibility (highlighted with a green box), Data privacy (highlighted with a red box), Advertising data, and Notifications. The main content area is titled 'Visibility of your profile & network' and lists various settings with their current status and a right-pointing arrow for editing:

Visibility of your profile & network	
Profile viewing options	Your name and headline →
Page visit visibility	On →
Edit your public profile	→
Who can see or download your email address	→
Who can see your connections	On →
Who can see members you follow	Anyone on LinkedIn →
Who can see your last name	→
Representing your organizations and interests	On →
Page owners exporting your data	Off →
Profile discovery and visibility off LinkedIn	→
Profile discovery using email address	Anyone →
Profile discovery using phone number	Everyone →
Blocked members	→

How to Protect Your Privacy on LinkedIn?

- Optout feeding your profile to AI.

← → ↻ 🔍 linkedin.com/mypreferences/d/categories/privacy

in

Settings

- Account preferences
- Sign in & security
- Visibility
- Data privacy**
- Advertising data
- Notifications

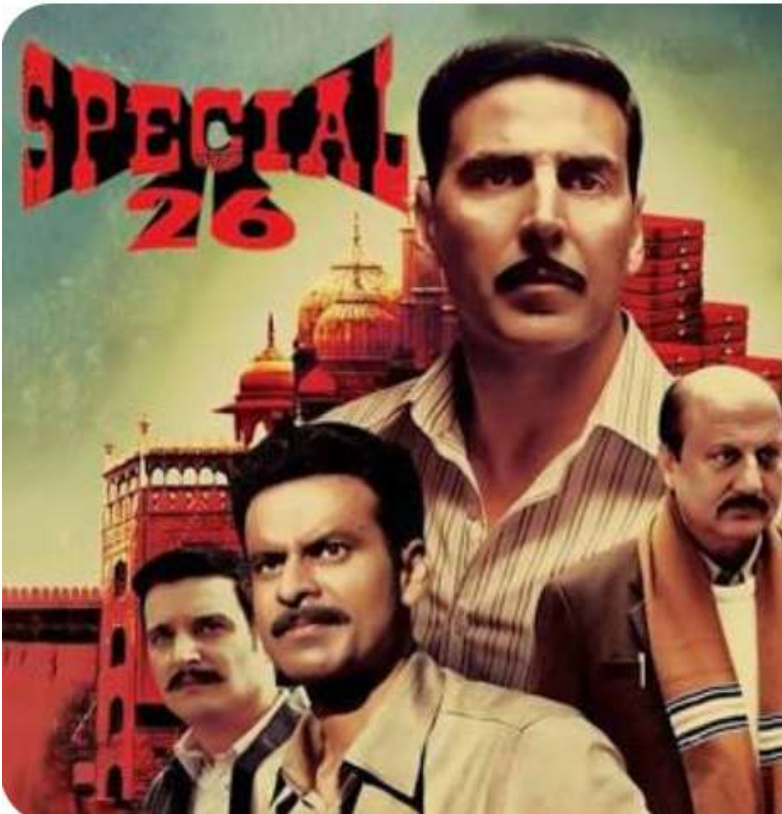
How LinkedIn uses your data

Manage your data and activity	→
Get a copy of your data	→
Search history	→
Personal demographic information	→
Social, economic, and workplace research	On →
Data for Generative AI Improvement	On →

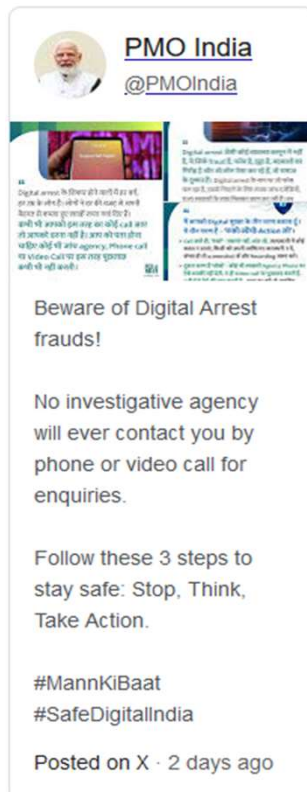
Who can reach you

Invitations to connect	→
Invitations from your network	→
Messages	→
Research invites	On →
LinkedIn promotions	→

Digital Arrest



Digital Arrest



- 'Mann Ki Baat': PM Modi cautions people against 'digital arrest' scam by fraudsters posing as law enforcement officials
- In just the first two months of 2025 (up to February 28), 17,718 digital-arrest / related cybercrime cases were reported, with a defrauded amount of ₹210.21 crore.
- In Karnataka, till 2025, 95 victims lost about ₹20.11 crore via digital-arrest scams.
- Stop, Think and Act.

After 3 days of 'digital arrest', Hyderabad retired doctor dies of heart attack; scammers sent texts after death

Mahesh Buddi / TNN / Updated: Sep 17, 2025, 10:52 IST

Share Print AA Select TOI as

A 76-year-old retired doctor in Hyderabad tragically died of a heart attack after enduring three days of relentless harassment by cyber fraudsters posing as law enforcement. The scammers extorted Rs 6.6 lakh from her under the guise of a 'digital arrest' and continued their threats even after her death. Police have registered ... [Read More](#)



79-yr-old Mumbai lawyer arrested, FIR filed again

Sep 15, 2025, 23:33 IST



SIM cards and bank accounts were fraudulently opened in the couple's names and were being used for money laundering and anti-national activities, said the officer privy to the probe. The scammers sent forged documents, including fake FIRs, arrest warrants, and other electronic papers that appeared authentic and informed the couple that they were under 'digital arrest' and could only prove their innocence by immediately transferring money to specified bank accounts. The case has been filed under various sections of BNS and IT Act.

Bengaluru Woman Recovers 14 Lakhs She Lost In Digital Arrest. Cops Acted Within 'Golden Hour'

The woman received a call from fraudsters posing as Mumbai Cyber Crime Police, claiming her bank account was linked to illegal transactions and threatened to arrest her unless she transferred money to an account for "verification."

Reported by: [Deepak Bopanna](#) [Bengaluru News](#) Sep 24, 2025 09:59 am IST

impersonated law enforcement officials. Police registered cases against three accused and managed to save Rs 5 lakh.

On Sept 3 the elderly lawyer received a threatening WhatsApp call at his Mumbra residence around 3 pm. The caller alleged that

Elderly man loses retirement savings of 1.8 crore in digital arrest fraud

Dwaipayan Ghosh / Sep 13, 2025, 23:58 IST

Share Print AA Preferred on G

NOTHING CALLED DIGITAL ARREST

> Beware of IVR calls. Cross-check with the actual source, the official website, for their authenticity
> There is nothing called digital arrest in Indian law

agencies never ask for personal information on phone or online
> Never click on links or download attachments/apps sent on SMS, emails or social media

Kolkata: A senior citizen from Netaji Nagar lost his entire retirement savings of Rs 1.8 crore to scamsters, who posed as cops from the Mumbai Crime Branch and threatened him with "digital arrest". A complaint was lodged on Friday.

According to police, the man, who used to work at



directed to a fraudulent website posing as a finance and investment services company.

A 38-year-old software engineer from Serilingampally lost over Rs 3.14 crore to cyber fraudsters who lured her into investing through a fake stock market portal, police said on Thursday.

According to the complaint filed with the Telangana State Cyber Security Bureau (TGCSB), the victim said she was added to a WhatsApp group named J7-Legacy Elite by one Sneha Sarda in July, according to TOI report. She was then

What is a digital arrest scam?

- A digital arrest scam is an online scam that defrauds victims of their hard-earned money.
- The scammers scare the victims and falsely accuse them of illegal activities.
- They later demand money and puts them under pressure for making the payment.



How digital arrest scam works?

- scammers pose as law enforcement officials, such as CBI agents, income tax officers, or customs agents, and initiate contact with victims via phone calls.
- Subsequently, they request that the victims switch to video communication through platforms like WhatsApp and Skype.
- The scammers then threaten the victims with a digital arrest warrant, citing various reasons such as financial misconduct, tax evasion, or other legal violations.
- In certain instances, these fraudsters create a police station-like set-up to further convince victims that the call is legitimate.
- Under the guise of “clearing their name”, “assisting with the investigation”, or “refundable security deposit/escrow account”, individuals are pressurized into transferring large sums of money to specified bank accounts or UPI IDs. Once the victims comply and make the payment, the scammers vanish, leaving the victims to face financial loss and potential identity theft.

How to avoid becoming a victim of digital arrest scam?

- Stop, Think and Act.
- Be suspicious of calls from fake officials claiming that you are in trouble. It is always important to remember that real law enforcement agency officials will never ask for payment or banking details.
- If you are suspicious about the call, verify their identity by directly contacting the relevant agency they are referring to. Try to be calm and do not panic.
- Avoid sharing personal information and never disclose sensitive personal or financial details over the phone or video calls, especially to unknown numbers.
- Remember, government agencies do not use platforms like WhatsApp or Skype for official communication.
- Call 1930

Know the number of connections issued in your name by logging in using your mobile number

- <https://tafcop.sancharsaathi.gov.in/telecomUser/>

भारत सरकार
GOVERNMENT OF INDIA

संचार मंत्रालय
MINISTRY OF COMMUNICATIONS

SKIP TO MAIN CONTENT

Q A ↩


दूरसंचार विभाग
DEPARTMENT OF TELECOMMUNICATIONS








SANCHAR SAATHI

ABOUT

CITIZEN CENTRIC SERVICES

KEEP YOURSELF AWARE

FAQs

IN SOCIAL MEDIA

AUTHORIZED LOGIN


TAF COP
 Know the number of connections issued in your name by logging in using your mobile number

9856816

requests

10 digit Mobile number



Queries?

▶ Thank You.

▶ Contact details

▶ sanjayadiwal@cdac.in

▶ +91-9916938713

